Robust Multimodal Road Extraction via Dual-Layer Evidential Fusion Networks for Remote Sensing

Hui Wang, *Graduate Student Member, IEEE*, Youxiang Huang, Yu Wang, *Member, IEEE*, Donglai Jiao, Hao Huang, *Member, IEEE*, Yun Lin, *Senior Member, IEEE*, and Guan Gui, *Fellow, IEEE*

***Abstract*— Accurate road network extraction from remote sensing images (RSIs) is essential for applications such as urban planning, map updates, and autonomous navigation. However, challenges such as complex backgrounds, varying spatial resolutions, and occlusions hinder traditional single-modality approaches, which often fail to capture comprehensive contextual information. To address these limitations, we propose the DEFNet, a novel dual-layer evidential fusion network for robust multimodal road extraction. The DEFNet features two key modules: cross-attention feature interaction (CAFI) and dual-layer evidential fusion (DEF). The CAFI module facilitates adaptive multimodal interaction at both pixel and superpixel levels, enhancing feature fusion while mitigating noise. The DEF module, leveraging the Dirichlet framework and Dempster–Shafer theory (DST), performs uncertainty-aware fusion, improving prediction reliability and robustness. Extensive experiments on multiple benchmark datasets demonstrate that the DEFNet consistently outperforms state-of-the-art methods in both accuracy and robustness, making it highly effective for multimodal road extraction in remote sensing applications. The codes can be downloaded from https://github.com/BeechburgPieStar/DEFNet**

***Index Terms*— Cross-attention feature interaction (CAFI), dual-layer evidential fusion (DEF), multimodal road extraction, remote sensing images (RSIs), uncertainty estimation.**

I. Introduction

EXTRACTING road networks from remote sensing images (RSIs) is essential for various applications, including urban planning, map updates, and autonomous navigation [1]. This task involves identifying and mapping road structures from aerial or satellite imagery, which is crucial for the accurate representation of transportation networks [2]. However, due to the diverse characteristics of RSIs, such as complex backgrounds, varying spatial resolutions, and occlusions from vegetation or buildings, road extraction remains a challenging problem [3].

Single-modality remote sensing road extraction primarily relies on deep learning techniques, particularly convolutional neural networks (CNNs) and graph neural networks (GNNs), which have significantly improved accuracy and connectivity [3], [4], [5], [6], [7]. The most commonly used methods are based on semantic segmentation for road extraction. Fully convolutional networks (FCNs) enable pixel-level classification for precise road extraction, with models such as the U-Net [8] and the ResUNet [9] effectively preserving spatial details via skip and residual connections. The D-LinkNet [10] expands the receptive field by using dilated convolutions, enhancing feature capture. To further improve the continuity of complex road shapes, the MSMDFF-Net [11] introduces multiscale and multidirectional feature fusion, achieving excellent performance in complex environments. Additionally, the lightweight BIR-Net [12] uses bilateral interaction modules, improving edge feature precision, and is suitable for resource-constrained environments.

Generative adversarial networks (GANs) have also been employed to enhance road boundary accuracy and connectivity. Mei et al. [13] proposed the CoANet, which uses adversarial learning to refine road boundaries, while Gao et al. [14] introduced the MCMCNet, incorporating multitask consistency constraints to preserve fine details in high-resolution images. The RADANet [15] employs attention mechanisms, further enhancing road extraction precision and connectivity in complex scenes. The AD-RoadNet [16] improves multiscale detail retention with an auxiliary-decoding strategy, and NIGAN [17] applies conditional GANs to enhance mountain road extraction performance.

GNNs offer a novel approach for modeling road topological structures. Zao et al. [18] proposed TopoRoad, which predicts vertex and edge relations to represent road networks as topological graphs, achieving high accuracy in road extraction. Wang et al. [19] developed TERNformer, which improves performance in complex urban environments by capturing both local connectivity and global context. Furthermore, the branch structure in TopoRoad [20] optimizes road connectivity by employing a minimum spanning tree algorithm.

Despite these advances, single-modality approaches face limitations, particularly in complex environments where roads are obscured or visually similar to other linear features like

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Hui Wang, Youxiang Huang, Yu Wang, Hao Huang, and Guan Gui are with the College of Telecommunications and Information Engineering, Nanjing University of Posts and Telecommunications, Nanjing 210003, China (e-mail: 1223014110@njupt.edu.cn; 1322018705@njupt.edu.cn; yuwang@njupt.edu.cn; huanghao@njupt.edu.cn; guiguan@njupt.edu.cn).

Donglai Jiao is with the College of Internet of Things, Nanjing University of Posts and Telecommunications, Nanjing 210003, China (e-mail: jiaodonglai@njupt.edu.cn).

Yun Lin is with the College of Information and Communication Engineering, Harbin Engineering University, Harbin 150000, China (e-mail: linyun@hrbeu.edu.cn).

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railways or rivers [21], [22]. To overcome the limitations of single-modality approaches, multimodal remote sensing road extraction has gained significant attention. By integrating different data sources such as RSIs [light detection and ranging (LiDAR)] and global positioning system (GPS) trajectories, multimodal methods leverage complementary information, improving road extraction accuracy, robustness, and connectivity, particularly in complex or occluded areas [23], [24], [25].

The fusion of imagery and GPS trajectory data is widely used in multimodal road extraction. Gao et al. [26] proposed a dual-attention dilated-LinkNet combined with floating vehicle trajectory data, significantly enhancing road extraction accuracy in complex environments. Liu et al. [27] integrated aerial imagery and crowdsourced GPS trajectories, optimizing road extraction by effectively processing trajectory data. Additionally, Bai et al. [28] presented an automatic road network generation approach by combining image and trajectory data, significantly improving network connectivity.

The integration of LiDAR and remote sensing imagery has shown considerable advantages in multimodal road extraction, especially in complex terrains and occluded areas. Parajuli et al. [24] demonstrated high road segmentation accuracy by combining aerial LiDAR and imagery, even in challenging environments. Zhang et al. [29] integrated GPS trajectories with high-resolution satellite imagery for precise road boundary detection in urban environments. Yang et al. [30] proposed DuARE, which combines aerial images and road networks, showing improved performance across varied terrains and complex scenes. Lightweight multimodal information propagation models have also gained attention, particularly for resource-constrained settings. Qiu et al. [31] developed an efficient framework that combines LiDAR and imagery for road extraction in low-resource environments. Filho et al. [32] further improved road network continuity with a multimodal deep learning model, enhancing contextual and contour information.

With the development of multimodal fusion techniques, the robustness and accuracy of road extraction in complex environments have significantly improved. For instance, NIGAN [17] applies conditional GANs to mountain road extraction, integrating remote sensing data for enhanced accuracy. The combination of LiDAR and imagery data has been particularly effective in addressing the challenges posed by complex terrains and occlusions, as demonstrated by Parajuli et al. [24]. The ability to fuse various data sources provides a powerful solution for road extraction, making it more adaptable to diverse real-world scenarios.

However, existing multimodal methods often have several limitations. They typically rely on fixed, manually designed feature fusion strategies, which may not adapt well to varying data quality and scene conditions. Furthermore, these methods often treat all modalities equally without adjusting their weights according to context-specific relevance, which can lead to degraded performance in cases where certain modalities are noisy or incomplete. Additionally, most current methods lack mechanisms for uncertainty estimation, resulting in unreliable predictions in complex scenarios.

To address these challenges, we propose a dual-layer evidential fusion network (DEFNet), a novel framework designed for multimodal road extraction that enables efficient information fusion and uncertainty handling. The DEFNet framework comprises two key modules: the cross-attention feature interaction (CAFI) module [33] and the dual-layer evidential fusion (DEF) module [34]. The CAFI module processes feature at both pixel and superpixel levels, allowing adaptive interaction between modalities by dynamically adjusting their importance. This results in more precise feature fusion, which enhances key information expression while mitigating noise. The DEF module integrates information at the basic probability assignment (BPA) and multimodal levels. First, DEF uses the Dirichlet framework to convert neural network outputs into Dirichlet distribution parameters for effective uncertainty estimation. Then, it applies the Dempster–Shafer theory (DST) to perform evidence fusion across BPA and multimodal levels [35], [36], [37], [38]. Specifically, the DEF module fuses two distinct evidence sources at the BPA level, generating preliminary belief and uncertainty information, and subsequently combines evidence from different modalities at the multimodal level. This dual-layer fusion strategy significantly enhances the model's robustness and prediction reliability in complex environments.

The main contributions of this article are highlighted below.

1) We propose the *DEFNet*, an innovative multimodal framework for remote sensing road extraction, which dynamically integrates multimodal information and performs uncertainty estimation, enhancing model accuracy and robustness.
2) We design the *CAFI* module, which enables feature interaction at both pixel and superpixel levels, adaptively integrating important information from each modality and thereby enhancing model performance in complex scenes.
3) We introduce the *DEF* module, which uses the Dirichlet framework and DST to perform dual-layer fusion at the BPA and multimodal levels, effectively improving prediction reliability and noise resilience.
4) We conduct extensive experiments on multiple datasets, and results demonstrate that the DEFNet outperforms existing methods in terms of accuracy and robustness, validating its effectiveness for multimodal road extraction tasks.

The rest of this article is organized as follows. Section II describes our proposed DEFNet in detail. Section III describes the implementation details and experimental results. Section IV concludes this work.

II. Our Proposed Method

In this section, we introduce the framework of our DEFNet and its two main components: the CAFI module and the DEF module.

A. Overall Framework

The overall framework of our DEFNet is shown in Fig. 1. In this study, we define the two input modalities, RSI and

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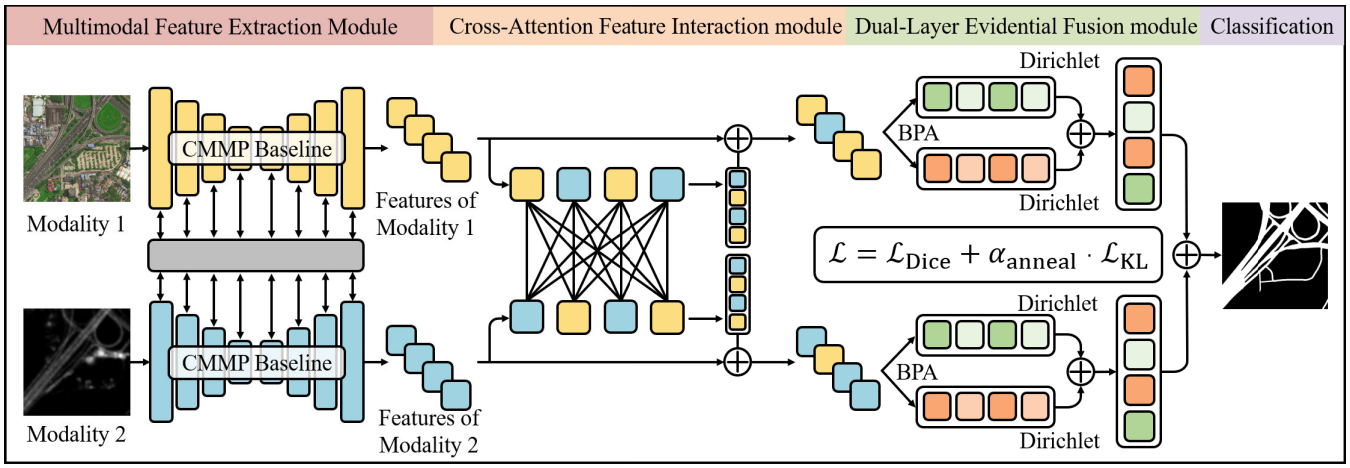


Fig. 1. Schematic illustration of the proposed DEFNet includes a multimodal feature extraction module, CAFI module, and the DEF module.

GPS trajectory (or LiDAR), as M_1 and M_2, respectively. Both modalities are processed through the CMMPNet baseline model to extract distinct features from each data source, which can be represented as

$$\mathbf{h}^{M_1}, \mathbf{h}^{M_2} = g\left(\mathbf{x}^{M_1}, \mathbf{x}^{M_2}\right) \tag{1}$$

where $\mathbf{h}^{M_1}$ and $\mathbf{h}^{M_2}$ denote the features extracted from modality M_1 and M_2, respectively, by the CMMPNet model.

To effectively integrate these modality-specific features, we introduce the CAFI module, enabling cross-modal information exchange and enhancing feature complementarity, which will be discussed in detail in Section II-B. Finally, the features are fused using the DEF module, which incorporates uncertainty estimation to refine the decision-making process and improve robustness, as further explained in Section II-C. The multimodal remote sensing road extraction is formulated as a multiple-input–single-output segmentation problem that aims to enhance road extraction accuracy by integrating data from multiple sensor modalities, thus improving the quality and consistency of road feature recognition.

Specifically, we consider two primary modalities: RSI, denoted as M_1 with $\mathbf{x}^{M_1} \in \mathcal{X}^{M_1}$, and GPS trajectory (or LiDAR) data, denoted as M_2 with $\mathbf{x}^{M_2} \in \mathcal{X}^{M_2}$, where $\mathcal{X}$ indicates the sample space for each modality. Each sample is paired with a binary label mask, $\mathbf{y} \in \mathcal{Y}$, where $\mathcal{Y}$ represents the label space, distinguishing road and nonroad regions. Consequently, the multimodal remote sensing road extraction problem, based on the maximum a posteriori (MAP) criterion, can be formulated as

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y} \in \mathcal{Y}} f\left(\mathbf{y} \mid \mathbf{x}^{M_1}, \mathbf{x}^{M_2}\right) \tag{2}$$

where $\hat{\mathbf{y}}$ denotes the predicted segmentation mask and $f(\cdot)$ represents the mapping function from the multimodal joint sample space $\mathcal{X} = \mathcal{X}^{M_1} \cup \mathcal{X}^{M_2}$ to the label space $\mathcal{Y}$.

The overall optimization objective of this model for multimodal remote sensing road extraction includes two main aspects: 1) minimizing the discrepancy between the predicted road maps and the ground-truth maps to improve extraction accuracy and 2) controlling the model's uncertainty and robustness by introducing regularization constraints that help mitigate noise while preserving essential multimodal information. The total optimization function can be formulated as

$$\min \frac{1}{N} \sum_{i=1}^{N} \left(\mathcal{L}_{\text{Dice}}\left(f\left(\mathbf{x}_i^{M_1}, \mathbf{x}_i^{M_2}\right), \mathbf{y}_i \right) + \alpha_{\text{anneal}} \cdot \mathcal{L}_{\text{KL}}(\alpha_i) \right) \tag{3}$$

where $\mathcal{L}_{\text{Dice}}(f(\mathbf{x}_i^{M_1}, \mathbf{x}_i^{M_2}), \mathbf{y}_i)$ represents the Dice Loss, maximizing overlap between predictions and ground truth. Here, $\mathbf{x}_i^{M_1}$ and $\mathbf{x}_i^{M_2}$ denote inputs from modality M_1 and modality M_2 for the ith sample. The Kullback–Leibler (KL) divergence term $\mathcal{L}_{\text{KL}}(\alpha_i)$ manages uncertainty, and $\alpha_{\text{anneal}} = \min(1.0, \text{epoch}/\text{max_epochs})$ scales its influence progressively. N represents the batch size. This optimization balances segmentation accuracy and uncertainty control, ensuring robust road extraction across multimodal inputs.

In summary, the DEFNet combines RSI and GPS trajectories (or LiDAR) to enhance road extraction accuracy and robustness. Through the CAFI module and the DEF module, the model integrates modality-specific features and manages uncertainty. The optimization balances Dice Loss for accuracy and KL divergence for uncertainty control, ensuring reliable and robust road extraction.

B. CAFI Module

To fully leverage the complementary information from different modalities, this module enhances each modality's feature representation through a cross-attention mechanism. Traditional single-modality fusion methods often struggle to capture complex relationships between different data sources. By introducing the CAFI module, as shown in Fig. 2, we enable deeper interaction between modalities, which improves the robustness and quality of the feature representations.

The features from modality 1 ($\mathbf{h}^{M_1}$) are processed using simple linear iterative clustering (SLIC) to generate superpixel-level representations, capturing high-level contextual information. Meanwhile, the features from modality 2 ($\mathbf{h}^{M_2}$) are kept at the pixel level to provide finer spatial details.

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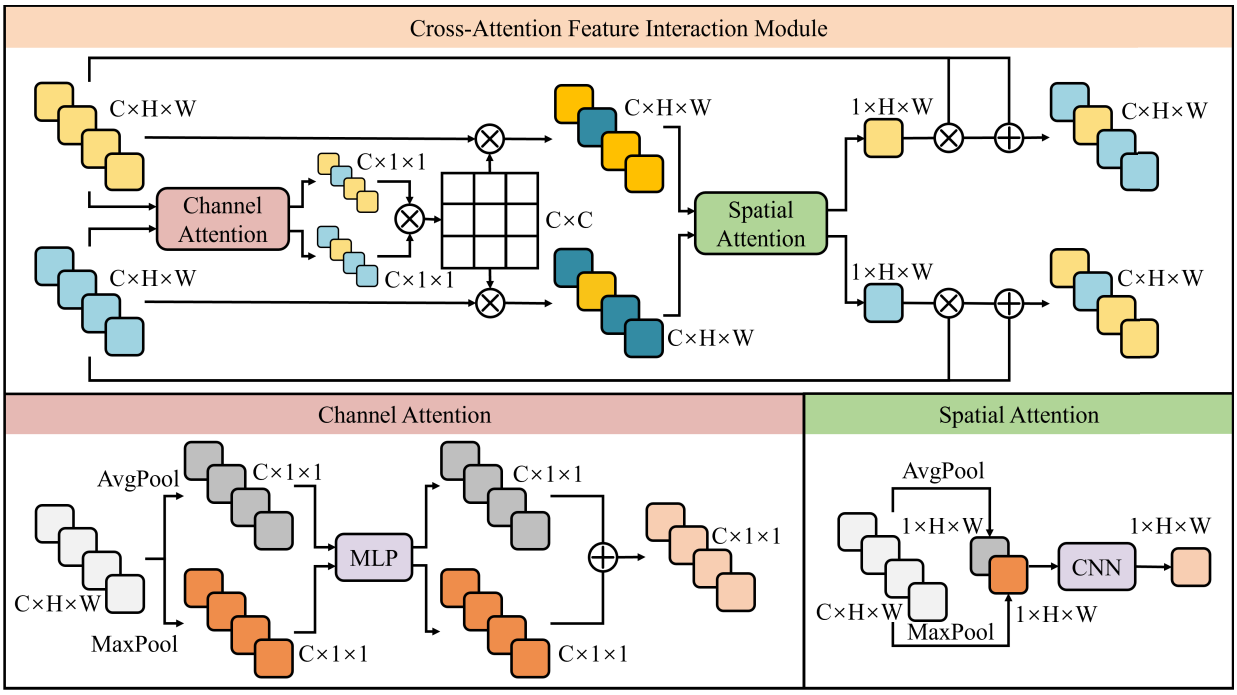


Fig. 2. Illustration of the CAFI module.

In the CAFI module, we enhance the feature representations from the two modalities by applying channel and spatial attention mechanisms. For each modality's input feature map, $\mathbf{H}^{M_1} \in \mathbb{R}^{C\times H\times W}$ and $\mathbf{H}^{M_2} \in \mathbb{R}^{C\times H\times W}$, we first generate channel descriptors through global average pooling and max-pooling, resulting in descriptors of shape $C \times 1 \times 1$. These channel descriptors are then processed by a shared two-layer fully connected network (MLP) to produce the channel attention weights $\mathbf{A}_{\text{ch}}^{M_1}$ and $\mathbf{A}_{\text{ch}}^{M_2}$

$$\mathbf{A}_{\text{ch}}^{M} = \sigma\left(\mathbf{W}_1 \cdot \text{AvgPool}\left(\mathbf{H}^{M}\right) + \mathbf{W}_2 \cdot \text{MaxPool}\left(\mathbf{H}^{M}\right)\right) \quad (4)$$

where σ is the rectified linear unit (ReLU) activation function and $\mathbf{W}_1$ and $\mathbf{W}_2$ are learnable weights. The channel attention weights $\mathbf{A}_{\text{ch}}^{M_1}$ and $\mathbf{A}_{\text{ch}}^{M_2}$ are applied to the features from modality 1 and modality 2, respectively, through element-wise multiplication to obtain enhanced channel features $\mathbf{H}_{\text{ch}}^{M_1}$ and $\mathbf{H}_{\text{ch}}^{M_2}$

$$\mathbf{H}_{\text{ch}}^{M_1} = \mathbf{A}_{\text{ch}}^{M_1} \cdot \mathbf{H}^{M_1}, \quad \mathbf{H}_{\text{ch}}^{M_2} = \mathbf{A}_{\text{ch}}^{M_2} \cdot \mathbf{H}^{M_2}. \quad (5)$$

Next, we further refine the feature representations using spatial attention. The channel descriptors of each modality are concatenated to generate spatial descriptors of shape $1 \times H \times W$. These descriptors are processed by a shared convolutional layer with a kernel size of $k \times k$, producing spatial attention weights $\mathbf{A}_{\text{sp}}^{M_1}$ and $\mathbf{A}_{\text{sp}}^{M_2}$

$$\mathbf{A}_{\text{sp}}^{M} = \sigma\left(f^{k\times k}\left(\text{Concat}\left(\text{AvgPool}\left(\mathbf{H}^{M}\right), \text{MaxPool}(\mathbf{H}^{M})\right)\right)\right) \quad (6)$$

where $f^{k\times k}$ represents the convolutional layer with kernel size $k \times k$. The spatial attention weights are then applied to each modality's features to obtain spatially enhanced features $\mathbf{H}_{\text{sp}}^{M_1}$ and $\mathbf{H}_{\text{sp}}^{M_2}$

$$\mathbf{H}_{\text{sp}}^{M_1} = \mathbf{A}_{\text{sp}}^{M_1} \cdot \mathbf{H}^{M_1}, \quad \mathbf{H}_{\text{sp}}^{M_2} = \mathbf{A}_{\text{sp}}^{M_2} \cdot \mathbf{H}^{M_2}. \quad (7)$$

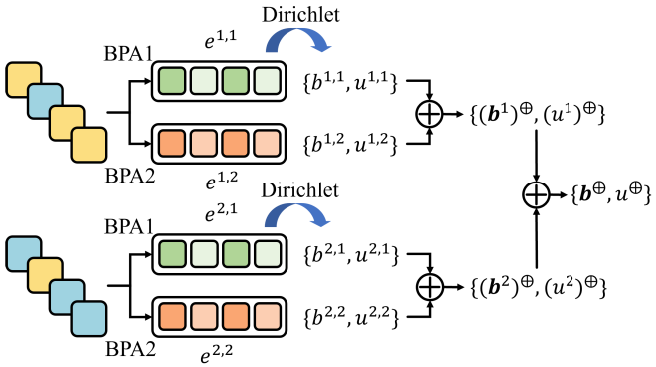


Fig. 3. Illustration of the DEF module.

Finally, we combine the original features with the attention-enhanced features through a residual connection to produce the output features for each modality

$$\mathbf{H}_{\text{out}}^{M} = \text{Softmax}\left(\mathbf{H}_{\text{sp}}^{M}\right) \cdot \mathbf{H}^{M} + \mathbf{H}^{M}. \quad (8)$$

These final output features $\mathbf{H}_{\text{out}}^{M_1}$ and $\mathbf{H}_{\text{out}}^{M_2}$ are used in subsequent layers, enabling effective multimodal feature fusion and representation.

C. DEF Module

The DEF module, as illustrated in Fig. 3, is designed to integrate complementary information from multiple modalities, aiming to achieve robust and reliable decision-making at the pixel level. This module is DST and combines multiple pieces of evidence and uncertainty distributions from each modality through a two-layer fusion process. Each modality's

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evidence is generated using two types of BPA methods: BPA1, which utilizes Softmax, and BPA2, which utilizes ReLU.

For modality 1, at a specific pixel location, two different pieces of evidence are generated using BPA1 and BPA2, denoted as $e_i^{1,1}$ and $e_i^{1,2}$, where the superscript $1,1$ represents evidence generated by BPA1 from modality 1, and $1,2$ represents evidence generated by BPA2 from modality 2. Specifically, the evidence $e_i^{1,1}$ generated by BPA1 is defined as

$$e_i^{1,1} = E^1 \cdot \frac{e^{x_i^1}}{\sum_j e^{x_j^1}} \tag{9}$$

where E^1 denotes the total evidence for modality 1, and x_i^1 is the network output for class i at this pixel location. The evidence $e_i^{1,2}$ generated by BPA2 is defined as

$$e_i^{1,2} = \max\left(0, x_i^1\right). \tag{10}$$

With these two pieces of evidence for modality 1, we calculate the corresponding belief values $b_i^{1,1}$ and $b_i^{1,2}$, and the uncertainties $u^{1,1}$ and $u^{1,2}$ from each evidence source, respectively. The total evidence sum for modality one is given by

$$S^1 = e_i^{1,1} + e_i^{1,2}. \tag{11}$$

Thus, the belief values for class i are computed as

$$b_i^{1,1} = \frac{e_i^{1,1}}{S^1}, \quad b_i^{1,2} = \frac{e_i^{1,2}}{S^1}. \tag{12}$$

The uncertainties for modality 1, derived from each evidence source, are then

$$u^{1,1} = \frac{K}{S^1}, \quad u^{1,2} = \frac{K}{S^1}. \tag{13}$$

In the first fusion layer, we fuse the two pieces of evidence within modality 1 to obtain the combined belief $(b_i^1)^{\oplus}$ and uncertainty $(u^1)^{\oplus}$. Using Dempster–Shafer combination rules, these are defined as

$$\left(b_i^1\right)^{\oplus} = \frac{1}{\kappa}\left(b_i^{1,1} \cdot b_i^{1,2} + b_i^{1,1} \cdot u^{1,2} + b_i^{1,2} \cdot u^{1,1}\right) \tag{14}$$

$$\left(u^1\right)^{\oplus} = \frac{1}{\kappa} u^{1,1} \cdot u^{1,2} \tag{15}$$

where κ is a normalization factor ensuring the sum of belief and uncertainty is 1. After completing the first fusion layer for modality 1, we apply the same process to modality 2 to obtain its fused belief $(b_i^2)^{\oplus}$ and uncertainty $(u^2)^{\oplus}$.

In the second fusion layer, we integrate the fused results from modality 1 and modality 2 to get the final belief $b_i^{\oplus}$ and uncertainty $u^{\oplus}$. The final fusion formulas are

$$b_i^{\oplus} = \frac{1}{\kappa}\left(\left(b_i^1\right)^{\oplus} \cdot \left(b_i^2\right)^{\oplus} + \left(b_i^1\right)^{\oplus} \cdot \left(u^2\right)^{\oplus} + \left(b_i^2\right)^{\oplus} \cdot \left(u^1\right)^{\oplus}\right) \tag{16}$$

$$u^{\oplus} = \frac{1}{\kappa}\left(u^1\right)^{\oplus} \cdot \left(u^2\right)^{\oplus}. \tag{17}$$

Finally, through this dual-layer fusion process, we obtain the cross-modal fused belief and uncertainty values, which are used to compute the final Dirichlet parameters. The fused Dirichlet parameter α_i is defined as

$$\alpha_i = S \cdot b_i^{\oplus} + 1 \tag{18}$$

where $S = K/u^{\oplus}$. This dual-layer fusion approach effectively integrates multimodal evidence, providing robust and comprehensive predictions for each class.

To prove that the fused uncertainty $u^{\oplus}$ in the second fusion layer is less than or equal to the minimum of the individual uncertainties u^1 and u^2, we start with the definition of $u^{\oplus}$

$$u^{\oplus} = \frac{u^1 \cdot u^2}{\sum_{i=1}^{K}\left(b_i^1 \cdot b_i^2 + u^1 \cdot b_i^2 + u^2 \cdot b_i^1\right) + u^1 \cdot u^2}. \tag{19}$$

By utilizing the normalization conditions $\sum_{i=1}^{K} b_i^1 + u^1 = 1$ and $\sum_{i=1}^{K} b_i^2 + u^2 = 1$, the denominator simplifies to

$$\sum_{i=1}^{K} b_i^1 \cdot b_i^2 + u^1 + u^2 - u^1 \cdot u^2. \tag{20}$$

Thus, the fused uncertainty can be rewritten as

$$u^{\oplus} = \frac{u^1 \cdot u^2}{\sum_{i=1}^{K} b_i^1 \cdot b_i^2 + u^1 + u^2 - u^1 \cdot u^2}. \tag{21}$$

To verify that $u^{\oplus} \leq u^1$, we can multiply both sides of the inequality by the denominator, yielding

$$u^1 \cdot u^2 \leq u^1\left(\sum_{i=1}^{K} b_i^1 \cdot b_i^2 + u^1 + u^2 - u^1 \cdot u^2\right). \tag{22}$$

Expanding the right-hand side and simplifying, we obtain

$$u^1 \cdot u^2 \leq u^1 \tag{23}$$

thus proving $u^{\oplus} \leq u^1$. By symmetry, we also find $u^{\oplus} \leq u^2$. Consequently, the fused uncertainty $u^{\oplus}$ is indeed less than or equal to the minimum of the individual uncertainties, confirming the effectiveness of the fusion in controlling uncertainty in the second layer.

D. Loss Function

For the multimodal remote sensing road extraction task, we combine Dice Loss and KL divergence. This combined loss is designed to enhance segmentation accuracy while promoting uncertainty estimation. An annealing coefficient gradually increases the weight of the KL term, allowing the model to focus on segmentation initially and progressively incorporate regularization. The total loss $\mathcal{L}$ is defined as

$$\mathcal{L} = \mathcal{L}_{\text{Dice}} + \alpha_{\text{anneal}} \cdot \mathcal{L}_{\text{KL}} \tag{24}$$

where $\mathcal{L}_{\text{Dice}}$ measures the overlap between predictions and ground truth and $\mathcal{L}_{\text{KL}}$ regularizes the predicted Dirichlet parameters toward a uniform distribution. The annealing coefficient α_{anneal} scales the KL term based on the training epoch. The Dice Loss is formulated as

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_c m_c \cdot y_c^{\text{true}} + \epsilon}{\sum_c m_c + \sum_c y_c^{\text{true}} + \epsilon} \tag{25}$$

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Algorithm 1 DEFNet for Multimodal Remote Sensing Road Extraction

Require: Input modalities: $\mathbf{x}^{M_1}$ (RSI), $\mathbf{x}^{M_2}$ (GPS Traj./LiDAR); training epochs N.
Ensure: Predicted segmentation mask $\hat{\mathbf{y}}$ with uncertainty.
Feature Extraction: Extract features $\mathbf{h}^{M_1}$ and $\mathbf{h}^{M_2}$ according to Eq. (1).
Cross-Attention Interaction: Enhance feature representations using cross-attention according to Section II-B.
Evidential Fusion:
for each pixel **do**
  Generate evidences $e_i^{1,1}$, $e_i^{1,2}$ for $\mathbf{h}^{M_1}$ and $e_i^{2,1}$, $e_i^{2,2}$ for $\mathbf{h}^{M_2}$ by Eqs. (9)–(10).
  Compute beliefs $b_i^{1,1}$, $b_i^{1,2}$ and uncertainties $u^{1,1}$, $u^{1,2}$ by Eqs. (12)–(13).
  Fuse intramodality beliefs and uncertainties to obtain $(b_i^1)^{\oplus}$, $(u^1)^{\oplus}$, $(b_i^2)^{\oplus}$, $(u^2)^{\oplus}$ by Eqs. (14)–(15).
end for
Cross-Modal Fusion: Fuse modality-specific results to obtain final belief $b_i^{\oplus}$ and uncertainty $u^{\oplus}$ according to Eqs. (16)–(17).
Loss Calculation: Calculate the total loss $\mathcal{L}$ according to Eq. (24).
Optimization: Minimize the total loss $\mathcal{L}$ over N epochs.
Prediction: Output the segmentation mask $\hat{\mathbf{y}}$ with uncertainty estimates.

where m_c and y_c^{true} are the predicted probability and ground-truth label for class c, respectively, and ϵ prevents division by zero.

The KL divergence term $\mathcal{L}_{\text{KL}}$ encourages uncertainty estimation by reducing the divergence between predicted parameters α and a target distribution β

$$\mathcal{L}_{\text{KL}} = \sum_{i=1}^{B}\left[\ln\frac{B(\beta)}{B(\alpha_i)} + \sum_{c=1}^{C}(\alpha_{i,c}-\beta_c)\,\psi(\alpha_{i,c})\right] \tag{26}$$

where $B(\cdot)$ is the Beta function, and $\psi(\cdot)$ is the digamma function. This regularization aligns the predicted distribution with the uniform target, aiding in uncertainty quantification. The annealing coefficient is defined as

$$\alpha_{\text{anneal}} = \min\left(1.0, \frac{\text{current_epoch}}{\text{max_epochs}}\right). \tag{27}$$

This coefficient gradually increases the influence of the KL Divergence, ensuring a balanced transition from segmentation accuracy to uncertainty regularization.

In summary, this composite loss function enables accurate segmentation with reliable uncertainty estimates, making it ideal for applications that require robust assessments. The detailed processing steps of our DEFNet framework are outlined in Algorithm 1.

III. Experimental Results and Analysis

A. Experimental Setup

1) Datasets: We evaluated our method on three publicly available multimodal road extraction datasets. The BJRoad dataset [39] contains 1350 high-resolution RSIs covering approximately 100 km^2 and around 50 million trajectories from 28 000 vehicles. Images are resized from 1024 × 1024 to 512 × 512 [27], and the dataset is split into 70% training, 10% validation, and 20% testing. The TLCGIS dataset [24] includes 5860 aerial and LiDAR image pairs (500 × 500 pixels, 0.5 feet per pixel) divided into 2640 training, 2400 testing, and 240 validation samples. The Porto dataset [40] covers approximately 209 km^2 in Porto, Portugal, with trajectories from 442 taxis. Following [40], we split the area into 6048 subimages (512 × 512), with one-fifth for testing and the rest for training.

2) Implementation Details: In this study, the DEFNet model is implemented on PyTorch 2.0.0 and compared with nine methods: four unimodal road segmentation-specific methods (UNet [8], DeepLabv3+ [41], D-LinkNet [10], and LFD-RoadSeg [42]) and five multimodal models (CMMPNet [27], CMNEXT [43], AsyFFNet [44], CMGFNet [45], and PAC-SCNet [46]) not exclusively developed for this task due to limited open-source code. Each method is tested ten times with average results and 95% confidence intervals reported. For the BJRoad dataset, data augmentations (flips, rotations, and random cropping) are applied to address limited data and complex scenes, with images resized as in [27]. No augmentations are used for the TLCGIS and Porto datasets to ensure fairness. Training details: 30 epochs (BJRoad), 50 epochs (others), batch size 4, learning rate 1e-4 with cosine annealing, and ResNet ImageNet pretrained weights. Experiments on an NVIDIA RTX 3090 GPU using AdamW optimizer minimize loss, enabling DEFNet to leverage multimodal information effectively for superior road segmentation performance across datasets.

3) Evaluation Metrics: In the task of multimodal remote sensing road extraction, we adopt commonly used evaluation metrics, including intersection over union (IoU), precision, recall, and $F1$-score. IoU assesses the overlap between the predicted road and the ground truth, serving as an essential metric for segmentation performance. Precision indicates the model's accuracy in predicting roads, while recall reflects the model's coverage of true road instances. The $F1$-score, as the harmonic mean of precision and recall, provides a balanced view of the model's overall performance. These metrics comprehensively evaluate the effectiveness of the model in remote sensing road extraction. Additionally, connectivity (CONN) [26] is introduced to evaluate the structural integrity of road networks by measuring the matching degree between the connected components of predictions and ground truth.

B. Performance Comparison

1) BJRoad Dataset: The proposed DEFNet demonstrates superior performance in key evaluation metrics on the BJRoad dataset, verifying its effectiveness in multimodal road extraction. As shown in Table I, the DEFNet achieves an IoU of 60.33%, significantly higher than the 59.42% of the CMMP-Net, the second-best multimodal method. Its $F1$-score reaches 74.47%, and the CONN is 73.98%, both ranking first among all compared methods. Among single-modal approaches, the D-LinkNet performs best with an IoU of 56.16%, yet its connectivity of 65.01% is significantly lower than its multimodal

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TABLE I
COMPARISON BETWEEN OUR DEFNET AND OTHER METHODS ON THE BJROAD DATASET (UNIT: %)

Type	Method	IoU	Precision	Recall	F1	CONN
Single-modal	Unet	55.74 ± 0.58	72.57 ± 0.95	71.83 ± 0.95	70.45 ± 0.53	58.20 ± 0.85
	DeepLabv3+	55.39 ± 0.51	73.69 ± 0.65	70.10 ± 1.11	69.95 ± 0.56	67.59 ± 0.80
	D-LinkNet	56.16 ± 0.99	73.38 ± 0.89	71.70 ± 1.13	70.83 ± 0.93	65.01 ± 0.75
	LFD-RoadSeg	52.85 ± 0.63	71.88 ± 0.85	67.83 ± 0.92	67.55 ± 0.27	52.64 ± 0.78
Multimodal	CMMPNet	59.42 ± 0.36	75.88 ± 0.53	74.53 ± 0.87	73.40 ± 0.38	73.58 ± 0.72
	CMNEXT	58.41 ± **0.27**	74.68 ± 0.96	73.31 ± 0.92	72.56 ± 0.25	51.75 ± 0.68
	AsyFFNet	54.90 ± 0.39	68.83 ± 0.98	66.08 ± 1.01	66.24 ± **0.19**	53.50 ± **0.60**
	CMGFNet	56.32 ± **0.27**	69.00 ± 0.81	75.99 ± 0.74	70.94 ± 0.23	50.58 ± 0.75
	PACSCNet	58.24 ± 0.46	**76.14** ± 0.73	72.05 ± 1.10	72.57 ± 0.39	72.09 ± 0.77
	Proposed	**60.33** ± 0.32	73.46 ± **0.29**	**77.83** ± **0.41**	**74.47** ± 0.28	**73.98** ± 0.70

TABLE II
COMPARISON BETWEEN OUR DEFNET AND OTHER METHODS ON THE TLCGIS DATASET (UNIT: %)

Type	Method	IoU	Precision	Recall	F1	CONN
Single-modal	Unet	68.20 ± 0.51	69.23 ± **0.20**	64.13 ± **0.10**	64.70 ± **0.19**	73.23 ± 0.90
	DeepLabv3+	73.08 ± 1.35	67.16 ± 0.63	65.74 ± 1.18	64.89 ± 0.87	72.72 ± 0.85
	D-LinkNet	75.98 ± **0.15**	67.61 ± 0.38	66.33 ± 0.45	65.87 ± 0.34	72.15 ± 0.92
	LFD-RoadSeg	59.77 ± 0.21	64.90 ± 0.82	58.00 ± 0.38	58.78 ± 0.27	70.76 ± 0.88
Multimodal	CMMPNet	70.06 ± 5.67	65.96 ± 1.98	64.06 ± 4.16	62.84 ± 3.86	73.16 ± 0.80
	CMNEXT	68.64 ± 0.37	65.04 ± 0.36	64.02 ± 0.99	62.71 ± 0.47	73.12 ± 0.75
	AsyFFNet	50.53 ± 2.73	57.20 ± 2.52	48.33 ± 2.44	48.64 ± 1.37	51.07 ± 0.78
	CMGFNet	64.17 ± 1.92	65.84 ± 0.78	67.80 ± 0.87	64.88 ± 0.61	61.94 ± 0.57
	PACSCNet	**78.18** ± 0.37	69.53 ± 0.51	67.79 ± 0.93	67.48 ± 0.55	73.39 ± **0.54**
	Proposed	76.15 ± 0.55	**70.91** ± 0.24	**69.45** ± 0.34	**68.69** ± 0.21	**73.75** ± 0.71

counterparts, indicating the limitation of single remote sensing imagery features in representing the integrity of the road network in complex scenes.

The CMMPNet achieves a connectivity of 73.58%, an improvement over DeepLabv3+, the best-performing single-modal method (67.59%), which reflects the positive effect of multisource data complementarity on road continuity to some extent. However, the performance advantages of some multimodal methods are not prominent on this dataset, mainly due to differences in design architectures and data processing strategies among different methods. The DEFNet further enhances performance through the CAFI module and DEF module, achieving an outstanding recall of 77.83%, a 3.3% increase over CMMPNet's 74.53%, which effectively strengthens the detection of roads in occluded or low-contrast areas. The model's standard deviation in IoU is ±0.32%, lower than most compared methods, reflecting its stability across different training iterations.

Visual comparisons (Fig. 4) align with quantitative results, clearly showing DEFNet's advantages in complex scenes. In road areas covered by building shadows, the CMMPNet produces fragmented predictions due to fixed feature fusion strategies [Fig. 4(c)], while the DEFNet generates continuous segmentation results that closely match ground-truth boundaries through cross-attention mechanisms at both pixel and superpixel levels [Fig. 4(d)]. In regions where roads meet similar-textured backgrounds, DEFNet's segmentation boundaries are smoother, significantly reducing misclassifications, which corresponds to its leading performance in the $F1$-score.

2) TLCGIS Dataset: On the TLCGIS dataset, the DEFNet demonstrates significant multimodal fusion advantages (Table II). While the PACSCNet leads with an IoU of 78.18%, the DEFNet achieves the best performance in precision, recall, $F1$-score, and CONN, reflecting more balanced segmentation performance. This IoU gap mainly results from the PACSCNet boosting IoU via aggressive boundary expansion, risking nonroad misclassifications. DEFNet's dual-layer evidential fusion balances intermodal uncertainties,

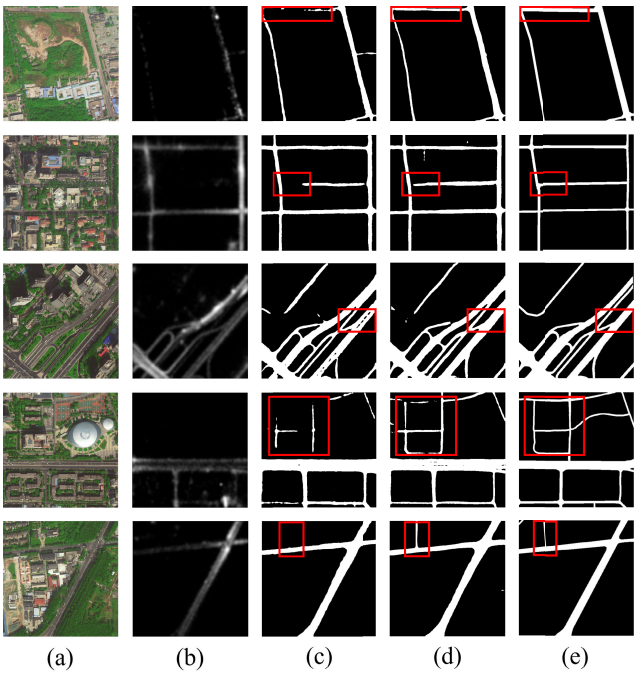


Fig. 4. Visual comparison of ours with the best competitors on the BJRoad dataset. We use the red box to highlight the regions with poor prediction results. (a) Image. (b) Trajectory. (c) CMMPNet. (d) Ours. (e) GT.

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TABLE III
COMPARISON BETWEEN OUR DEFNET AND OTHER METHODS ON THE PORTO DATASET (UNIT: %)

Type	Method	IoU	Precision	Recall	F1	CONN
Single-modal	Unet	68.90 ± **0.14**	73.92 ± **0.22**	74.48 ± **0.30**	72.93 ± 0.15	89.43 ± 0.92
	DeepLabv3+	65.71 ± 0.20	71.63 ± 0.73	71.77 ± 1.05	70.62 ± 0.12	92.13 ± 0.88
	D-LinkNet	68.46 ± 0.18	73.23 ± 0.34	73.65 ± 0.41	72.60 ± **0.10**	94.60 ± 0.85
	LFD-RoadSeg	60.27 ± 0.41	69.50 ± 0.54	67.40 ± 0.45	66.82 ± 0.48	76.51 ± 0.85
Multimodal	CMMPNet	69.04 ± 0.20	74.20 ± 0.50	73.66 ± 0.54	72.95 ± 0.19	93.87 ± 0.78
	CMNEXT	65.26 ± 0.24	71.11 ± 0.30	72.17 ± 0.46	70.48 ± 0.22	94.29 ± 0.75
	AsyFFNet	53.88 ± 0.55	62.69 ± 0.74	63.59 ± 0.75	61.55 ± 0.26	65.36 ± 0.91
	CMGFNet	60.66 ± 0.59	69.36 ± 0.71	73.62 ± 1.52	70.06 ± 0.64	84.49 ± **0.72**
	PACSCNet	66.92 ± 0.55	70.82 ± 0.37	76.02 ± 0.71	71.89 ± 0.44	91.24 ± 0.78
	Proposed	**69.60** ± 0.26	**74.41** ± 0.48	**76.36** ± 0.41	**74.34** ± 0.17	**94.86** ± 0.81

TABLE IV
MODEL COMPLEXITY OF DIFFERENT MODELS

Method	Single-modal				Multimodal					
	Unet	DeepLabv3+	D-LinkNet	LFD-RoadSeg	CMMPNet	CMNEXT	AsyFFNet	CMGFNet	PACSCNet	Proposed
FLOPs (G)	160.51	173.38	33.6	4.73	74.33	37.7	81.05	79.3	92.5	74.79
Params (M)	17.26	39.76	31.1	0.93	84.99	26.65	0.27	43.68	98.7	85.01

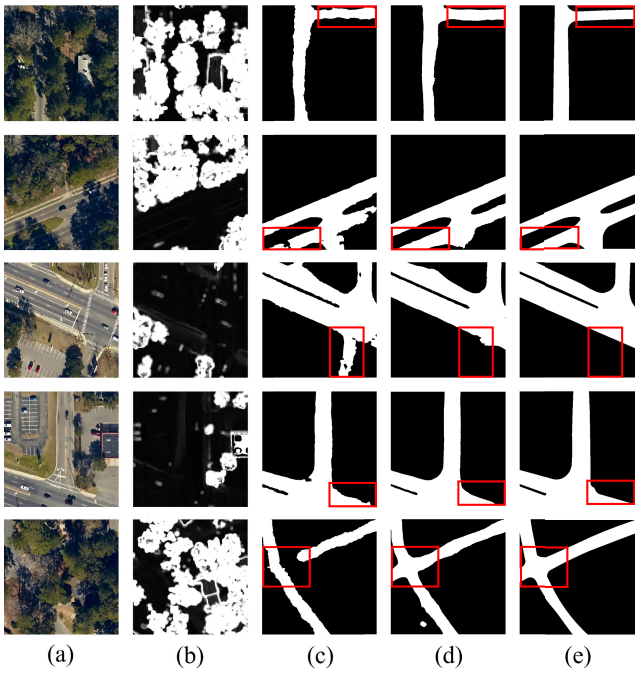


Fig. 5. Visual comparison of ours with the best competitors on the TLCGIS dataset. We use the red box to highlight the regions with poor prediction results. (a) Image. (b) LiDAR. (c) PACSCNet. (d) Ours. (e) GT.

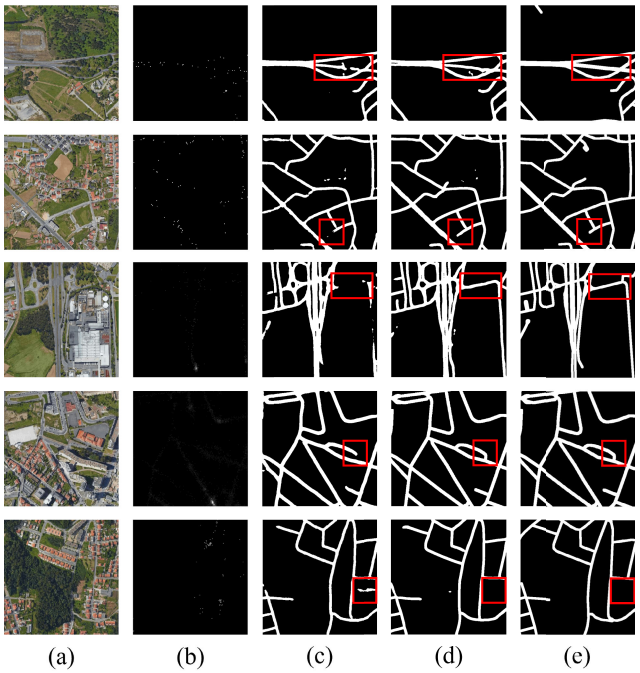


Fig. 6. Visual comparison of ours with the best competitors on the Porto dataset. We use the red box to highlight the regions with poor prediction results. (a) Image. (b) Trajectory. (c) CMMPNet. (d) Ours. (e) GT.

suppressing noise for accuracy in complex terrains, thus trading off IoU.

Among single-modal methods, the D-LinkNet achieves an IoU of 75.98% but shows lower connectivity (72.15%) in complex terrains due to reliance on single imagery features. The DEFNet dynamically integrates imagery and LiDAR data via cross-attention and evidential fusion, improving recall in vegetated areas and reducing omissions. Its precision exceeds the PACSCNet by 1.38%, indicating stronger class discrimination.

Visual results (Fig. 5) show that the DEFNet generates more continuous road boundaries matching ground truth in complex intersections and vegetated regions, whereas the PACSCNet exhibits fragmentation or ambiguity due to fixed fusion [Fig. 5(c) and (d)]. The DEFNet high CONN metric is visually validated by all components connected to the road, confirming the effectiveness of multimodal fusion for the extraction of complex streets in the scene.

3) Porto Dataset: As shown in Table III, D-LinkNet, a single-modal approach, achieves a high CONN of 94.60%, but its reliance on unimodal remote sensing imagery results in a lower IoU. In contrast, the multimodal CMMPNet achieves an IoU of 69.04%. However, some multimodal methods fail to outperform their single-modal counterparts on this dataset.

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TABLE V
MODAL ABLATION EXPERIMENTS ON THREE DATASETS (UNIT: %)

Modal	BJRoad					TLCGIS					Porto				
	IoU	Precision	Recall	F1	CONN	IoU	Precision	Recall	F1	CONN	IoU	Precision	Recall	F1	CONN
IMG	56.97	70.58	76.74	71.49	68.59	74.56	69.91	67.70	66.86	72.81	67.52	71.85	74.10	71.75	89.03
Traj./LiDAR	50.82	65.30	69.85	65.89	69.11	61.55	62.20	59.17	58.03	68.41	36.61	44.47	43.41	42.70	67.12
IMG+Traj./LiDAR	**60.33**	**73.46**	**77.83**	**74.47**	**73.98**	**76.15**	**70.91**	**69.45**	**68.69**	**73.75**	**69.60**	**74.41**	**76.36**	**74.34**	**94.86**

This is because the single-modal methods used in this study are specifically designed for road segmentation tasks and can fully utilize prior knowledge in the field of road segmentation. In contrast, due to the lack of dedicated open-source code for multimodal road extraction, the multimodal methods were not originally developed solely for this task. Their model architectures and parameter settings may not be fully adapted to road segmentation tasks, thus limiting their performance.

Our proposed DEFNet, specifically tailored for multimodal road extraction, enhances recall to 76.36% through dynamic integration of image spectral and trajectory spatial features via cross-attention, combined with DEF for uncertainty handling, effectively reducing omissions in narrow and intersecting road segments. Across multiple datasets, the DEFNet achieves the best balance of metrics, highlighting its superiority in leveraging multimodal information for accurate road segmentation.

Visual results (Fig. 6) show that the DEFNet generates more continuous segmentation boundaries that match the truth of the ground at road intersections and parallel segments, with its high CONN reflected in complete components connected to the road, markedly better than the fragmented predictions of the baselines due to fixed fusion strategies. The model's low standard deviations across metrics indicate stable generalization, providing a robust solution for multimodal road extraction in complex urban environments.

4) Model Complexity: Model complexity analysis reveals significant differences in computational cost and parameter scale among methods (Table IV). In single-modal models, Unet and DeepLabv3+ exhibit high complexity with 160.51G and 173.38G FLOPs, and 17.26M and 39.76M parameters, respectively. The D-LinkNet reduces FLOPs to 33.6G through a lightweight design but maintains a parameter size of 31.1M. LFD-RoadSeg, as an extremely lightweight model, features only 4.73G FLOPs and 0.93M parameters, though its performance is limited by feature extraction capacity.

Among multimodal models, the AsyFFNet stands out with the smallest parameter scale (0.27M) but still requires 81.05G FLOPs, indicating high computational cost. The PACSCNet has the largest parameters (98.7M) and FLOPs (92.5G), reflecting a complex architecture. The proposed DEFNet balances complexity with 74.79G FLOPs and 85.01M parameters, slightly higher than CMMPNet (74.33G, 84.99M) but significantly lower than the PACSCNet. This demonstrates that the DEFNet enables efficient multimodal fusion without excessive computational overhead, making it suitable for real-world applications that require a balance between efficiency and performance.

C. Ablation Studies

The ablation study results in Tables V and VI, along with the visualizations in Fig. 7, highlight the impact of different modalities and modules on model performance across three datasets.

Modal ablation experiments show that multimodal inputs consistently outperform single-modal counterparts across all datasets, confirming the necessity of heterogeneous data fusion. In the BJRoad dataset, single-modal imagery (IoU = 56.97%) is hindered by complex backgrounds, while sparse trajectories (IoU = 50.82%) lack structural details. Their fusion, leveraging trajectory structural priors and image visual features, boosts IoU to 60.33%, with recall and connectivity reaching 77.83% and 73.98%, respectively, enhancing road continuity in low-contrast areas. In TLCGIS, LiDAR alone (IoU = 61.55%) struggles without spectral information, but fusion with imagery utilizes 3-D terrain data to increase IoU to 76.15%, improving boundary detection in vegetated regions. In Porto, trajectories alone (IoU = 36.61%) fail due to missing geometry, while imagery alone (CONN = 89.03%) lacks dense network clarity; fusion clarifies spatial relationships via trajectories, raising connectivity to 94.86%, demonstrating cross-modal synergy in complex scenes.

Module ablation shows CAFI and DEF gains vary with modal noise and consistency. In BJRoad's noisy trajectories, DEF boosts recall by 4.22% and connectivity by 1.12% via noise suppression, while CAFI improves $F1$ by 0.55% through dynamic feature weighting; their synergy raises IoU by 1.36%. In TLCGIS, DEF alone achieves 76.83% IoU by fusing LiDAR's high-certainty evidence, slightly outperforming dual modules (76.15%), though CAFI optimizes precision (70.91%) and $F1$ (68.69%) via dynamic terrain classification. In Porto's clean data, dual modules enhance recall by 2.79% and connectivity by 1.0%, refining intersection boundaries through cross-modal attention with subtle quantitative gains.

The visual comparison in Fig. 7 further illustrates these effects on the BJRoad dataset. Key differences are marked by red boxes. When using only one modality or module, fragmented or less accurate road segments appear, especially in complex intersections. The complete model generates smoother, more continuous road segments, closely matching the ground truth, thus confirming the benefit of multimodal and modular integration.

In summary, the ablation study reveals that the combination of IMG and Traj./LiDAR modalities, along with both DEF and CAFI modules, significantly enhances model performance by improving road connectivity, boundary accuracy.

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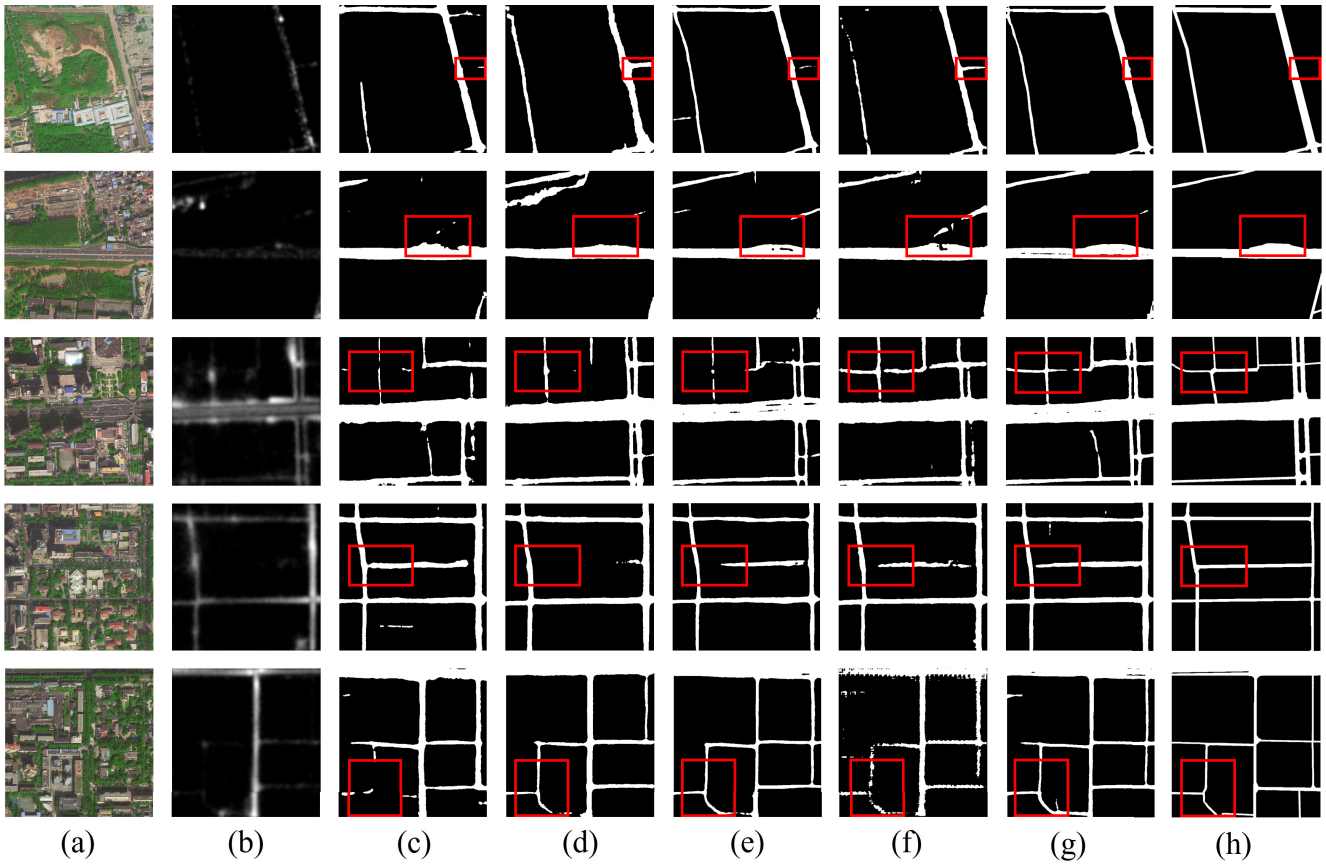


Fig. 7. Visual comparison of different ablation experiments on the BJRoad dataset. We use the red box to highlight the regions with poor prediction results. (a) Image. (b) Trajectory/LiDAR. (c) Image-only. (d) Trajectory/LiDAR-only. (e) DEF-only. (f) CAFI-only. (g) Proposed. (h) GT.

TABLE VI
MODULE ABLATION EXPERIMENTS ON THREE DATASETS

DEF	CAFI	BJRoad					TLCGIS					Porto				
		IoU	Precision	Recall	F1	CONN	IoU	Precision	Recall	F1	CONN	IoU	Precision	Recall	F1	CONN
×	×	58.97	**77.53**	72.94	72.90	72.58	73.86	66.79	66.59	65.51	73.16	69.52	**74.75**	73.57	73.33	93.87
✓	×	59.58	73.74	77.16	73.68	73.70	**76.83**	69.12	**69.91**	68.21	73.47	69.40	72.56	76.30	74.31	93.89
×	✓	59.35	77.46	72.97	73.45	73.57	76.27	67.61	66.72	65.91	73.22	69.59	73.12	73.67	73.25	94.14
✓	✓	**60.33**	73.46	**77.83**	**74.47**	**73.98**	76.15	**70.91**	69.45	**68.69**	**73.75**	**69.60**	74.41	**76.36**	**74.34**	**94.86**

IV. CONCLUSION

This study reveals the interaction between data characteristics and model design in multimodal road extraction through ablation experiments. Heterogeneous data fusion significantly improves extraction performance in complex scenarios. The hierarchical framework of cross-attention and evidential fusion modules enables adaptive optimization, which is verified by both visual analysis and quantitative metrics.

However, limitations remain. The research lacks exploration of emerging modalities like SAR, affecting the model's generalization in special environments. The framework's stability under imbalanced data or novel modal combinations requires verification, and some modules have high computational costs, unsuitable for large-scale data processing and edge device deployment.

Future research could focus on emerging modal fusion, algorithm lightweighting, and cross-modal prior mining. Improving the cross-attention mechanism for special modal data, developing dynamic fusion strategies, and constructing joint learning frameworks will enhance semantic understanding. Additionally, exploring transfer-learning techniques across different modalities and optimizing the model architecture for better resource utilization will be crucial steps in driving the technology toward greater robustness and efficiency.

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